



Girraween High School

2018

TRIAL HIGHER SCHOOL CERTIFICATE
EXAMINATION

Mathematics Extension 2

General Instructions

- Reading time: 5 minutes
- Working time: 3 Hours
- Write using a black or blue pen
- Board approved calculators may be used
- Laminated reference sheets are provided
- Answer multiple choice questions by completely colouring in the appropriate circle on your multiple choice answer sheet on the front page of your answer booklet.
- In questions 11-16 start all questions on a separate page in your answer booklet and show all relevant mathematical reasoning and/or calculations.

Total Marks: 100

Section 1 (Pages 2– 5) **10 Marks**

- Attempt Q1 - Q10
- Allow about 15 minutes for this section

Section 2 (Pages 5-13) **90 marks**

- Attempt Q11 – Q16
- Allow about 2 hours and 45 minutes for this section

Section 1 (10 marks)

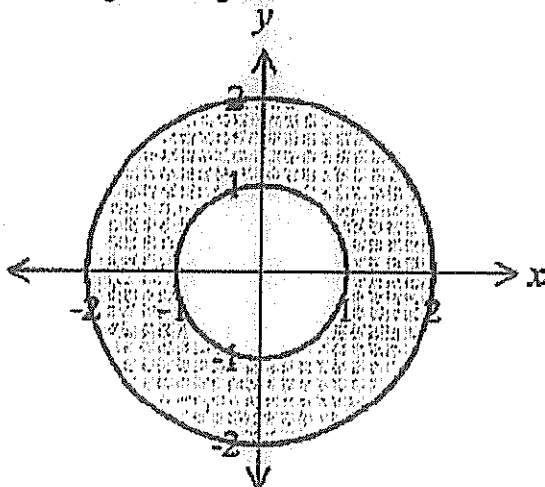
Allow about 15 minutes for this section.

Fill in the appropriate circle in your answer booklet.

1. Given that $z = 1 + i$, what is the value of z^8 ?

- (A) -16 (B) -8 (C) 8 (D) 16

2. Consider the Argand diagram below.



Which inequality could define the shaded area?

- (A) $0 \leq |z| \leq 2$
(B) $1 \leq |z| \leq 2$
(C) $0 \leq |z-1| \leq 2$
(D) $1 \leq |z-1| \leq 2$

3. The equation $x^4 + px + q = 0$, where $p \neq 0$ and $q \neq 0$ has roots α, β, γ and δ . What is the value of $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$?

- (A) $-4q$ (B) $p^2 - 2q$ (C) $p^4 - 2q$ (D) p^4

4. When $x^y = e$ is implicitly differentiated with respect to x , the result for $\frac{dy}{dx}$ is

- (A) $\frac{-y}{x \log_e x}$ (B) $\frac{y}{x \log_e x}$ (C) $\frac{-x \log_e x}{y}$ (D) $\frac{x \log_e x}{y}$

5. Which of the following is an expression for $\int_0^{\frac{\pi}{2}} \frac{1}{1 + \sin x} dx$

after the substitution $t = \tan \frac{x}{2}$?

(A) $\int_0^1 \frac{1}{1 + 2t} dt$ (B) $\int_0^1 \frac{2}{1 + 2t} dt$ (C) $\int_0^1 \frac{1}{(1 + t)^2} dt$ (D) $\int_0^1 \frac{2}{(1 + t)^2} dt$

6. What are the equations of the directrices of the hyperbola with equation

$$\frac{x^2}{144} - \frac{y^2}{25} = 1 ?$$

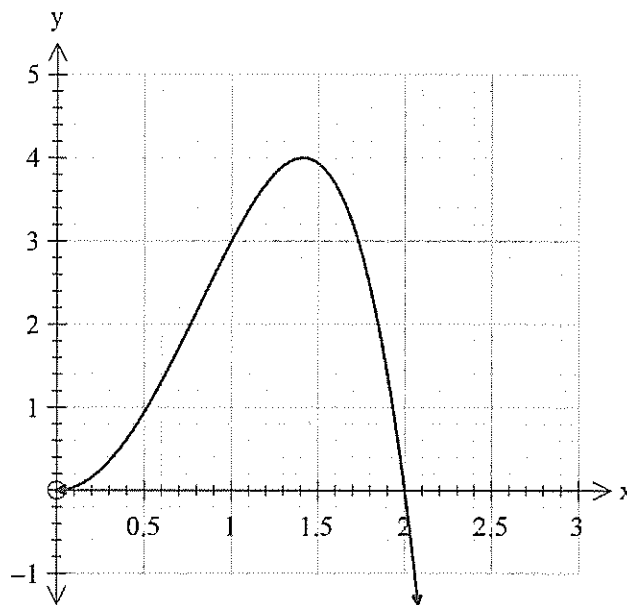
(A) $x = \pm \frac{13}{144}$ (B) $x = \pm \frac{13}{25}$ (C) $x = \pm \frac{25}{13}$ (D) $x = \pm \frac{144}{13}$

7. The region enclosed by $y = \sin x$, $y = 0$ and $x = \frac{\pi}{2}$ is rotated about the y -axis

to produce a solid. What is the volume of this solid using the method of cylindrical shells?

(A) π cubic units (B) $\frac{\pi}{2}$ cubic units (C) $\frac{3\pi}{2}$ cubic units (D) 2π cubic units

8. The graph of $y = 4x^2 - x^4$ is given below.



The region in the first quadrant bounded by the curve $y = 4x^2 - x^4$ and the x -axis between $x = 0$ and $x = 2$ is rotated about the y -axis.

Which of the following is an expression for the volume, V , of the solid formed?

(A) $V = 2\pi \int_0^4 \sqrt{4-y} \, dy$

(B) $V = 4\pi \int_0^4 \sqrt{4-y} \, dy$

(C) $V = 8\pi \int_0^4 \sqrt{4-y} \, dy$

(D) $V = 16\pi \int_0^4 \sqrt{4-y} \, dy$

9. A wheel of radius 2 metres rotates at 1200 revolutions per minute.

What is the tangential velocity of a point on the wheel?

(A) 40 m/s

(B) 80 m/s

(C) 251 m/s

(D) 260 m/s

10. A particle of mass m is moving horizontally in a straight line. Its motion is opposed by a force of magnitude $2m(v + v^2)$ Newtons when its speed is v m/s. At time t seconds the particle has a displacement of x metres from a fixed point O on the line and velocity v m/s.
Which of the following is an expression for x in terms of v ?

(A) $-\frac{1}{2} \int \frac{1}{1+v} dv$

(B) $-\frac{1}{2} \int \frac{1}{v(1+v)} dv$

(C) $\frac{1}{2} \int \frac{1}{1+v} dv$

(D) $\frac{1}{2} \int \frac{1}{v(1+v)} dv$

Section 2

Question 11 (15 marks)

- a. $z = p + 2i$, where p is a real number, and $w = 1 - 2i$ represent two complex numbers.

(i) Find $\frac{z}{w}$ in the form $a + ib$, where a and b are real numbers. [2]

(ii) Given that $\left| \frac{z}{w} \right| = 13$, find all possible values of p . [2]

b. $z = 1 - \sqrt{3}i$

(i) Find the values of $|z|$ and $\arg z$. [2]

(ii) Find the exact value of z^6 . [2]

c. (i) On an Argand diagram, sketch the locus of z represented by $|z - 3| = 3$. [2]

(ii) Explain why $\arg(z - 3) = 2 \arg z$. [2]

d. If $2 + i$ is a root of $P(x) = x^4 - 6x^3 + 9x^2 + 6x - 20$, resolve $P(x)$ into irreducible factors over the complex field. [3]

Question 12 (15 marks)

a. Find

(i) $\int x e^{-x} dx$ [2]

(ii) $\int_0^{\frac{\pi}{2}} \frac{1}{2 - \cos x + 2 \sin x} dx$ [3]

(iii) $\int_0^{\frac{\pi}{6}} \sec^3 2\theta d\theta$ [3]

b. (i) Find real numbers a, b, c and d such that:

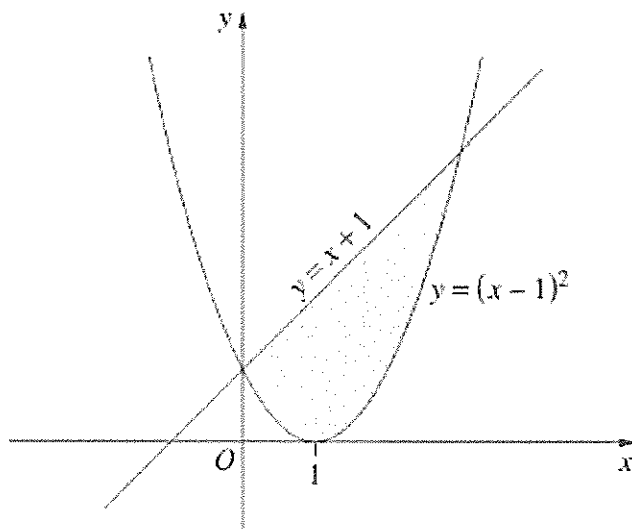
$$\frac{5x^3 - 3x^2 + 2x - 1}{x^2(x^2 + 1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{cx + d}{x^2 + 1}$$
 [2]

(ii) Hence find $\int \frac{5x^3 - 3x^2 + 2x - 1}{x^2(x^2 + 1)} dx$ [2]

c. The diagram shows the region enclosed by the curves $y = x + 1$ and $y = (x - 1)^2$.

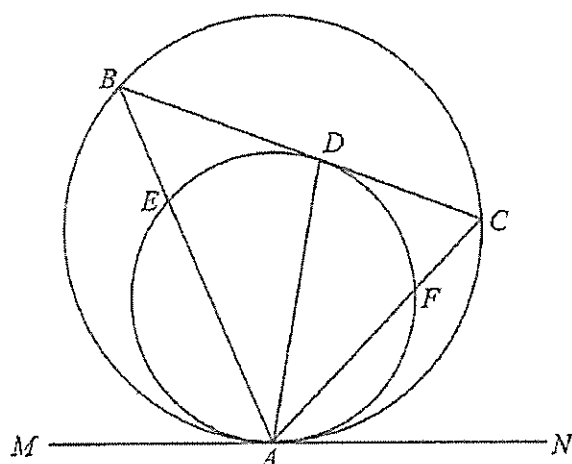
The region is rotated about the y -axis.

Find the volume of the solid using the method of cylindrical shells. [3]



Question 13 (15 marks)

a.



In the diagram, MAN is the common tangent to two circles touching internally at A.

B and C are two points on the larger circle such that BC is a tangent to the smaller circle with point of contact at D. AB and AC cut the smaller circle at E and F respectively.

(i) Copy or trace the into your answer booklet.

(ii) Show that AD bisects $\angle BAC$ [4]

b. An ellipse has the equation $\frac{x^2}{100} + \frac{y^2}{75} = 1$.

(i) Sketch the curve, showing the coordinates of the foci and the equations of the directrices. [2]

(ii) Find the equation of the normal to the ellipse at the point $P\left(5, 7\frac{1}{2}\right)$. [2]

(iii) Find the equation of the circle that is tangential to the ellipse at P and $Q\left(5, -7\frac{1}{2}\right)$. [3]

c. (i) Show that the tangent to the curve $xy = c^2$ at $T\left(ct, \frac{c}{t}\right)$ is given by

$$x + t^2y = 2ct \quad [2]$$

(ii) The tangent cuts the x and y axes at A and B respectively.

Prove that T is the centre of the circle that passes through O , A and B

where O is the origin. [2]

Question 14 (15 marks)

a. (i) Prove that $\int_0^a f(x) \, dx = \int_0^a f(a-x) \, dx$. [2]

(ii) Hence, find the value of $\int_0^2 x(2-x)^5 \, dx$. [2]

b. Given that $I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} \, dx$, where n is a positive integer,

(i) Show that $I_{2n+1} = \frac{1}{2} e - n I_{2n-1}$. [3]

(ii) Hence, or otherwise, evaluate $\int_0^1 x^5 e^{x^2} \, dx$. [3]

c. (i) Use the binomial theorem to expand $(\cos \theta + i \sin \theta)^3$. [1]

(ii) Use De Moivre's Theorem and your result from (i) to prove that

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta. \quad [2]$$

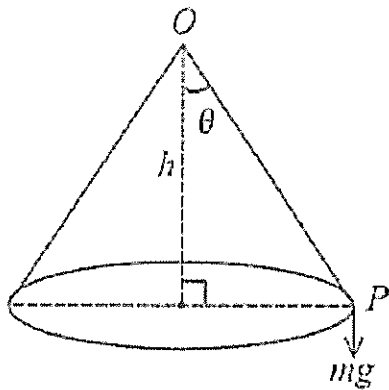
(iii) Hence, or otherwise, find the smallest positive solution of

$$4\cos^3 \theta - 3\cos \theta = 1 \quad [2]$$

Question 15 (15 marks)

a. A mass of m kg at P is suspended by a light inextensible string from point O .

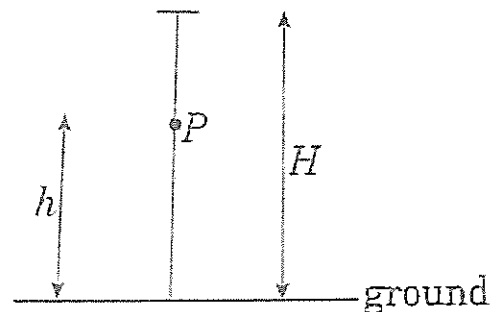
It describes a circle with a constant speed in a horizontal plane whose vertical distance below O is h metres.



(i) Show that $\omega = \sqrt{\frac{g}{h}}$. [2]

(ii) What is the period of the motion? [1]

b.



From a point on the ground an object of mass m kg is projected vertically upward with an initial speed of u . The object reaches a maximum height of H before falling back to the ground. The resistance to motion is equal to mkv^2 and g is the acceleration due to gravity.

(i) Show that $H = \frac{1}{2k} \ln \left(\frac{g + ku^2}{g} \right)$. [2]

(ii) P is a point at height h above the point of projection.

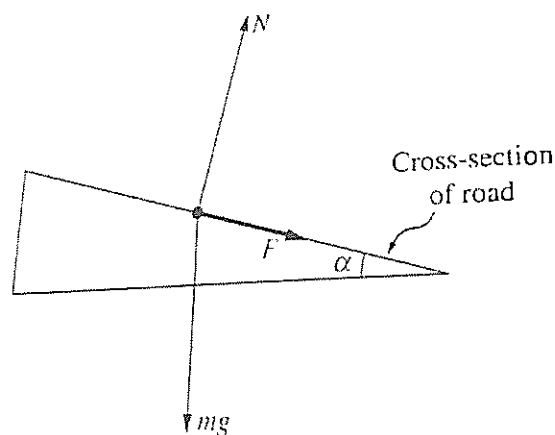
Let V be the speed of the object at P on its upward path when $x = h$.

Show that $h = \frac{1}{2k} \ln \left(\frac{g + ku^2}{g + kV^2} \right)$. [2]

(iii) During the downward path of the object it passes through P with half the speed of when it was first at P .

Show that $V = \sqrt{\frac{3g}{k}}$. [3]

c.



A road contains a bend that is part of a circle of radius, r . At the bend, the road is banked at an angle of α to the horizontal. A car travels around the bend at constant speed, v . Assume that the car is represented by a point of mass m , and that the forces acting on the car are the gravitational force mg , a sideways frictional force F (acting down the road) and a normal reaction N to the road.

(i) By resolving the horizontal and vertical components of force, find expressions for

$$F \cos \alpha \text{ and } F \sin \alpha . \quad [2]$$

(ii) Show that $F = \frac{m(v^2 - gr \tan \alpha)}{r} \cos \alpha . \quad [2]$

(iii) Suppose that the radius of the bend is 200 metres and that the road is banked to allow cars to travel at 100 km/h with no sideways friction force. Take $g = 9.8 \text{ ms}^{-2}$. Find the value of α . [1]

Question 16 (15 marks)

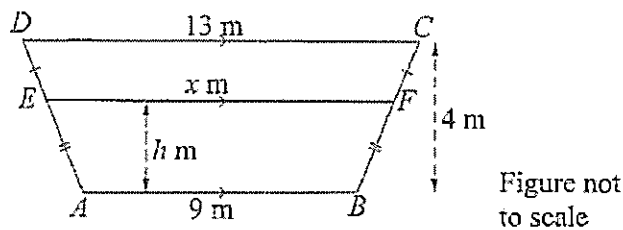
- a. $P(x)$ is a polynomial of degree 5 such that $P(x) - 1$ is divisible by $(x - 1)^3$ and $P(x)$ itself is divisible by x^3 . Derive an expression for $P(x)$.

[3]

- b. (i) The diagram below shows a trapezium $ABCD$ whose parallel sides AB and DC are 9 metres and 13 metres respectively.

The distance between these sides is 4 metres and $AD = BC$.

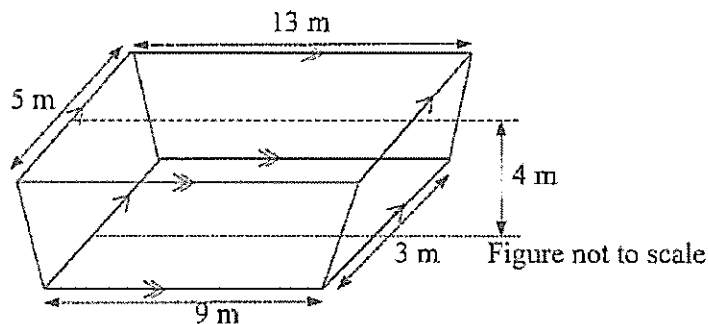
EF is parallel to AB and the distance between them is h metres.



Show that $EF = (9 + h)$ metres.

[2]

- (ii) The trench in the diagram below has a rectangular base with sides 9 metres and 3 metres. Its top is also rectangular with dimensions 13 metres and 5 metres. The trench has a depth of 4 metres and each of its four side faces is an isosceles trapezium.



Find the volume of the trench.

[4]

Question 16 continues on Page 13

- c. (i) Show that if $y = mx + k$ is a tangent to the hyperbola $xy = c^2$,
then $k^2 + 4mc^2 = 0$. [3]
- (ii) Hence, find the equations of the tangents from the point $(-1, -3)$
to the rectangular hyperbola $xy = 4$ and find their points of contact. [3]

End of Examination



①

GHS 2018 TRIAL NSC MATHEMATICS EXT. 2 SOLUTIONS
MC.

1. $z = 1 + i$, $z^8 = ?$
 $= \sqrt{2} \operatorname{cis} \frac{\pi}{4}$

$$\begin{aligned} z^8 &= (\sqrt{2} \operatorname{cis} \frac{\pi}{4})^8 \\ &= 2^4 (\cos 2\pi + i \sin 2\pi) \\ &= 16 \end{aligned}$$

D

2. $1 \leq |z| \leq 2$

B

3. $x^4 + px + q = 0$

$$\alpha^4 + p\alpha + q = 0$$

$$\beta^4 + p\beta + q = 0$$

$$\gamma^4 + p\gamma + q = 0$$

$$\delta^4 + p\delta + q = 0$$

$$\alpha^4 + \beta^4 + \gamma^4 + \delta^4 = -4p(\alpha + \beta + \gamma + \delta) - 4q$$

$$= -4p(0) - 4q$$

$$= -4q$$

A

4. $x^y = e$

$$\log x^y = \log e$$

$$y \log x = 1$$

$$y \cdot \frac{1}{x} + \log x \frac{dy}{dx} = 0$$

$$\frac{y}{x} + \log x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-y}{x \log x}$$

A

①



(2)

$$5. \int_0^{\pi/2} \frac{1}{1 + \sin x} dx$$

$$t = \tan \frac{x}{2}$$

$$= \int_0^1 \frac{1}{1 + \frac{2t}{1+t^2}} \cdot \frac{2dt}{1+t^2}$$

$$\frac{dt}{dx} = \frac{1}{2} \sec^2 \frac{x}{2}$$

$$= \int_0^1 \frac{2}{(1+t^2+2t)} dt$$

$$dx = \frac{2dt}{\sec^2 \frac{x}{2}}$$

$$= \int_0^1 \frac{2}{(1+t)^2} dt$$

$$dx = \frac{2dt}{1+t^2}$$

$$\text{When } x = \frac{\pi}{2}, t = 1$$

$$\text{When } x = 0, t = 0$$

D

$$6. \frac{x^2}{144} - \frac{y^2}{25} = 1$$

$$b^2 = a^2(e^2 - 1)$$

$$e^2 - 1 = \frac{25}{144}$$

$$e^2 = \frac{169}{144}$$

$$e = \frac{13}{12}$$

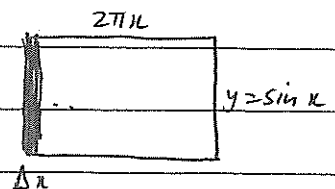
$$\text{Directrices : } x = \pm \frac{a}{e}$$

$$= \pm \frac{144}{13}$$

D

$$7. V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^{\pi/2} 2\pi x \sin x \Delta x$$

$$= 2\pi \int_0^{\pi/2} x \sin x dx$$



$$= 2\pi \left[(x \cos x)^{\pi/2} + \int_0^{\pi/2} \cos x dx \right]$$

$$= 2\pi \left[\sin x \right]_0^{\pi/2}$$

$$= 2\pi$$

D



(3)

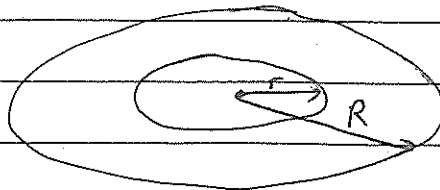
8. $y = 4x^2 - x^4$

$$(x^4 - 4x^2 + 4) = -y + 4$$

$$(x^2 - 2)^2 = 4 - y$$

$$x^2 - 2 = \pm \sqrt{4 - y}$$

$$x^2 = 2 + \sqrt{4 - y} = R^2; \quad x^2 = 2 - \sqrt{4 - y} = r^2$$



$$A = \pi(R^2 - r^2)$$

$$= \pi(2 + \sqrt{4 - y} - (2 - \sqrt{4 - y}))$$

$$= \pi(2\sqrt{4 - y})$$

$$= 2\pi\sqrt{4 - y}$$

$$\Delta V = 2\pi\sqrt{4 - y} \Delta y$$

$$V = \lim_{\Delta y \rightarrow 0} \sum_{y=0}^4 2\pi\sqrt{4 - y} \Delta y$$

$$V = 2\pi \int_0^4 \sqrt{4 - y} dy$$

A

9. $\omega = 1200 \text{ rpm}$

$$= \frac{1200 \times 2\pi}{60}$$

$$= 40\pi \text{ radians/sec}$$

$$v = r\omega$$

$$= 2 \times 40\pi$$

$$= 80\pi$$

$$\approx 251 \text{ m/s}$$

C

10. $m\ddot{x} = -2m(v + v^2)$

$$\ddot{x} = -2(v + v^2)$$

$$v \frac{dv}{dx} = -2(v + v^2)$$

$$\frac{dx}{dv} = \frac{-1}{2(1+v)}$$

$$x = -\frac{1}{2} \int \frac{1}{1+v} dv$$

A



(4)

Question 11

a) $z = p + 2i$; $w = 1 - 2i$

i) $\frac{z}{w} = \frac{p+2i}{1-2i} \times \frac{1+2i}{1+2i}$

$$= \frac{p + 2pi + 2i - 4}{1+4}$$

$$= \frac{p-4}{5} + \frac{2(p+1)}{5} i$$

(2)

ii) $\left| \frac{z}{w} \right| = 13$

$$\frac{|z|}{|w|} = 13$$

$$\frac{p^2+4}{5} = 169$$

$$p^2 = 841$$

$$p = \pm \sqrt{841} = \pm 29$$

(2)

b) $z = 1 - \sqrt{3}i$

i) $|z| = \sqrt{1+3} = 2$

$$\arg z = -\frac{\pi}{3}$$

(2)

ii) $z = 2 \operatorname{cis} -\frac{\pi}{3}$

$$z^6 = (2 \operatorname{cis} -\frac{\pi}{3})^6$$

$$= 2^6 (\cos -2\pi + i \sin -2\pi)$$

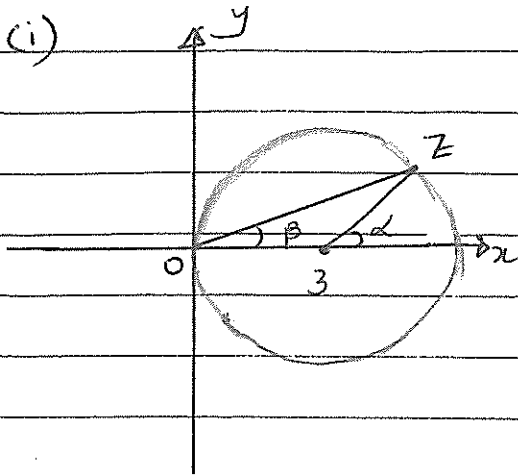
$$= 64$$

(2)



(5)

C. (i)



(2)

$$\therefore \alpha = \arg(Z-3) \quad ; \quad \beta = \arg Z$$

$\alpha = 2\beta$ ($\angle$ at the centre is twice the $\angle$ at the circumference subtended on same arc)

$$\therefore \arg(Z-3) = 2\arg Z$$

(2)

d. $2+i$ is a root

$\therefore 2-i$ is also a root ($P(x)$ has real coefficients)

$\therefore x^2 - 4x + 5$ is a factor.

$$\begin{array}{r}
 x^2 - 2x - 4 \\
 x^2 - 4x + 5 \overline{) x^4 - 6x^3 + 9x^2 + 6x - 20} \\
 \underline{x^4 - 4x^3 + 5x^2} \\
 -2x^3 + 4x^2 + 6x \\
 \underline{-2x^3 + 8x^2 - 10x} \\
 -4x^2 + 16x - 20 \\
 \underline{-4x^2 + 16x - 20} \\
 0
 \end{array}$$

$$P(x) = (x^2 - 4x + 5)(x^2 - 2x - 4)$$

Roots of $P(x)$ are $2+i, 2-i, 1+\sqrt{5}, 1-\sqrt{5}$

Factors : $(x - (2+i)), (x - (2-i)), (x - (1+\sqrt{5})), (x - (1-\sqrt{5}))$

(3)



(6)

Question 12

a) i)

$$\int x e^{-x} dx$$

$$u = x \quad v = -e^{-x}$$

$$u' = 1 \quad v' = e^{-x}$$

$$= -x e^{-x} - \int -e^{-x} dx$$

$$= -x e^{-x} + \int e^{-x} dx$$

$$= -x e^{-x} - e^{-x} + C$$

(2)

ii)

$$\int_0^{\pi/2} \frac{1}{2 - \cos x + 2 \sin x} dx$$

$$\text{let } t = \tan \frac{x}{2}$$

$$dt = \frac{1}{2} \sec^2 \frac{x}{2} dx$$

$$= \int \frac{1}{2 - \left(\frac{1-t^2}{1+t^2}\right) + 2\left(\frac{2t}{1+t^2}\right)} \cdot \frac{2dt}{1+t^2}$$

$$= \frac{1}{2} (1+t^2) dx$$

$$dx = \frac{2dt}{1+t^2}$$

$$= \int_0^1 \frac{2 dt}{2(1+t^2) - (1-t^2) + 4t}$$

$$\text{When } x=0, t=0$$

$$x = \pi/2 \quad t=1$$

$$= \int_0^1 \frac{2}{1 + 3t^2 + 4t} dt$$

$$= \int_0^1 \frac{2}{(3t+1)(t+1)} dt$$

$$\frac{A}{3t+1} + \frac{B}{t+1} = \frac{2}{(3t+1)(t+1)}$$

$$= \int_0^1 \left(\frac{3}{3t+1} - \frac{1}{t+1} \right) dt$$

$$A(t+1) + B(3t+1) = 2$$

$$\text{Substitute } t = -1$$

$$-2B = 2 \Rightarrow B = -1$$

$$\text{Substitute } t = -\frac{1}{3}$$

$$\frac{2}{3}A = 2$$

$$A = 3$$

$$= \left[\log(3t+1) - \log(t+1) \right]_0^1$$

$$= (\log 4 - \log 2) - (\log 1 - \log 1)$$

$$= \log 2$$

(3)



(7)

$$\text{iii) } \int_0^{\pi/6} \sec^3 2\theta \, d\theta$$

$$= \int_0^{\pi/6} \sec^2 2\theta \cdot \sec 2\theta \, d\theta$$

$$= \left[\frac{\tan 2\theta \cdot \sec 2\theta}{2} \right]_0^{\pi/6} - \int_0^{\pi/6} \sec 2\theta \tan^2 2\theta \, d\theta$$

$$= \left(\frac{\tan \frac{\pi}{3} \cdot \sec \frac{\pi}{3}}{2} \right) - \int_0^{\pi/6} \sec 2\theta (\sec^2 2\theta - 1) \, d\theta$$

$$= \frac{\sqrt{3}}{2} \cdot 2 - \int_0^{\pi/6} (\sec^3 2\theta + \sec 2\theta) \, d\theta$$

$$= \sqrt{3} - \int_0^{\pi/6} \sec^3 2\theta \, d\theta + \int_0^{\pi/6} \sec 2\theta \, d\theta$$

$$2 \int_0^{\pi/6} \sec^3 2\theta \, d\theta = \sqrt{3} + \int_0^{\pi/6} \sec 2\theta \, d\theta$$

$$= \sqrt{3} + \int_0^{\pi/6} \frac{\sec 2\theta (\sec 2\theta + \tan 2\theta)}{(\sec 2\theta + \tan 2\theta)} \, d\theta$$

$$= \sqrt{3} + \frac{1}{2} \int_0^{\pi/6} \frac{2 \sec^2 2\theta + 2 \sec 2\theta \tan 2\theta}{\sec 2\theta + \tan 2\theta} \, d\theta$$

$$= \sqrt{3} + \frac{1}{2} \ln [\sec 2\theta + \tan 2\theta]_0^{\pi/6}$$

$$= \sqrt{3} + \frac{1}{2} \ln (2 + \sqrt{3})$$

$$\therefore \int_0^{\pi/6} \sec^3 2\theta \, d\theta = \frac{\sqrt{3}}{2} + \frac{1}{4} \ln (2 + \sqrt{3})$$

$$\text{let } u = \sec 2\theta$$

$$= \frac{1}{\cos 2\theta}$$

$$= (\cos 2\theta)^{-1}$$

$$u' = -1 (\cos 2\theta)^{-2} \cdot -2 \sin 2\theta$$

$$= \frac{2 \sin 2\theta}{\cos^2 2\theta}$$

$$= 2 \sec 2\theta \cdot \tan 2\theta$$

$$v' = \sec^2 2\theta$$

$$v = \frac{\tan 2\theta}{2}$$

(3)



(8)

$$b) i) \frac{5x^3 - 3x^2 + 2x - 1}{x^2(x^2 + 1)} = \frac{a}{x} + \frac{b}{x^2} + \frac{cx + d}{x^2 + 1}$$

$$5x^3 - 3x^2 + 2x - 1 = ax(x^2 + 1) + b(x^2 + 1) + (cx + d)x^2$$

$$= (a+c)x^3 + (b+d)x^2 + ax + b$$

comparing :

Constant term $\Rightarrow b = -1$

Coefficients of x : $a = 2$

Coefficients of x^2 : $b + d = -3$

$$d = -2$$

Coefficients of x^3 : $a + c = 5$

$$c = 3$$

$$\therefore a = 2, b = -1, c = 3, d = -2$$

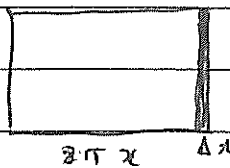
(2)

$$ii) \int \frac{5x^3 - 3x^2 + 2x - 1}{x^2(x^2 + 1)} dx = \int \left(\frac{2}{x} - \frac{1}{x^2} + \frac{3x - 2}{x^2 + 1} \right) dx$$

$$= \int \left(\frac{2}{x} - \frac{1}{x^2} + \frac{3x}{x^2 + 1} - \frac{2}{x^2 + 1} \right) dx$$

$$= 2 \log x + \frac{1}{x} + \frac{3}{2} \log(x^2 + 1) - 2 \tan^{-1} x + C$$

c)



$$y_2 - y_1 =$$

$$(x+1) - (x-1)^2$$

$$= -x^2 + 3x$$

$$\Delta V = 2\pi x (-x^2 + 3x) \Delta x$$

$$V = \lim_{\Delta x \rightarrow 0} \sum_{x=0}^3 2\pi (-x^3 + 3x^2) \Delta x$$

$$= 2\pi \int_0^3 (-x^3 + 3x^2) dx$$

$$= 2\pi \left[-\frac{x^4}{4} + x^3 \right]_0^3$$

$$= \frac{27\pi}{2}$$

Points of intersection :

$$(x-1)^2 = x+1$$

$$x^2 - 2x + 1 = x + 1$$

$$x^2 - 3x = 0$$

$$x = 0, 3$$

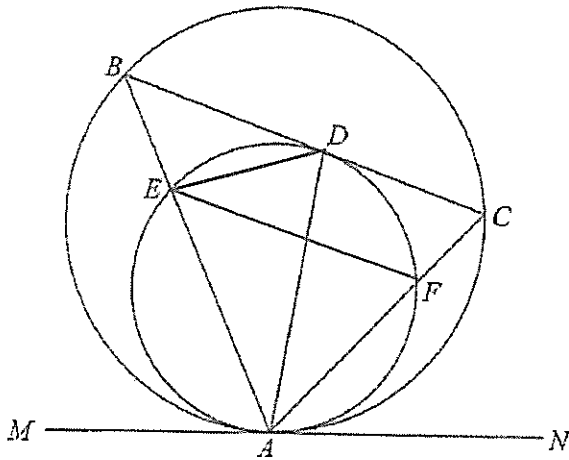
(3)



(9)

Question 13

4) i)



ii) Construct EF, ED.

$$\angle ABC = \angle NAC \text{ (}\angle \text{ in the alternate segment)}$$

$$\angle AEF = \angle NAC \text{ (}\angle \text{ in the alternate segment)}$$

$$\therefore EF \parallel BC \text{ (corresponding } \angle \text{s on transversal AB)}$$

$$\angle DEF = \angle BDE \text{ (alternate } \angle \text{s, } EF \parallel BC)$$

$$\angle DEF = \angle DAF \text{ (}\angle \text{s subtended by arc DF at the circumference of circle AEF)}$$

$$\angle BDE = \angle DAE \text{ (}\angle \text{ in alternate segment in circle AEF)}$$

$$\therefore \angle DAF = \angle DAE$$

$$\therefore AD \text{ bisects } \angle BAC.$$

$$b) \frac{x^2}{100} + \frac{y^2}{75} = 1$$

$$a=10, b=5\sqrt{3}$$

$$i) b^2 = a^2(1-e^2)$$

$$75 = 100(1-e^2)$$

$$e^2 = \frac{1}{4}$$

$$e = \frac{1}{2}$$

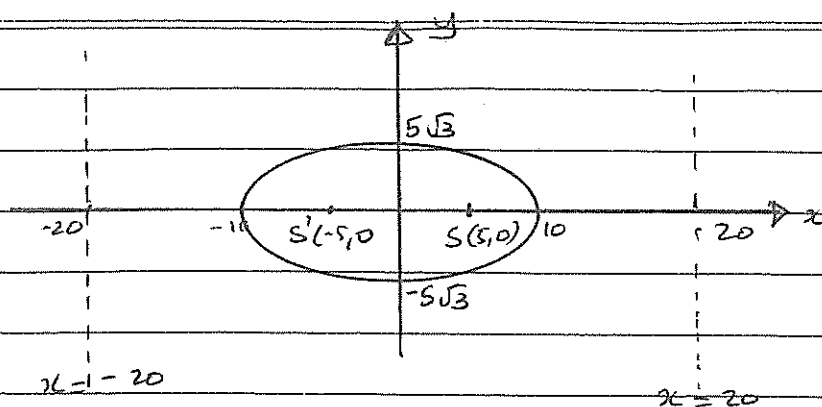
$$\text{Foci : } (\pm ae, 0)$$

$$\text{Foci : } (\pm 5, 0) ; \text{ Directrices : } x = \pm \frac{a}{e}$$

$$\text{Directrices : } x = \pm 20$$



(16)



$$ii) \frac{x^2}{100} + \frac{y^2}{75} = 1$$

Differentiating implicitly,

$$\frac{2x}{100} + \frac{2y}{75} \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{3x}{4y}$$

At $(5, 7\frac{1}{2})$

$$m_{\text{tangent}} = \frac{-3(5)}{4(7.5)} = -\frac{1}{2}$$

$$\therefore m_{\text{normal}} = 2$$

$$E_{\text{normal}} : y - 7.5 = 2(x - 5)$$

$$y = 2x - 2.5$$

iii) Circle and ellipse have common tangent at P

$\therefore$ Normal at P also at right $\angle$ s to circle at P

$\therefore$ Normal at P passes through centre, C, of circle.

$$\therefore C \text{ lies on } y = 2x - 2.5 \quad \text{--- (1)}$$

$$\text{Tangent at Q } (5, -7.5) \text{ has gradient, } m = \frac{-3(5)}{4(-7.5)} = \frac{1}{2}$$

$$\therefore m_{\text{normal at Q}} = -2$$

$$E_{\text{normal at Q}} : y = -2x + 2.5$$

$\Rightarrow$ also passes through centre, C, of circle

$$\therefore C \text{ lies on line } y = -2x + 2.5 \quad \text{--- (2)}$$



(11)

Solving ① & ② simultaneously

$$2x - 2 \cdot 5 = -2x + 2 \cdot 5$$

$$4x = 5$$

$$x = \frac{5}{4}, y = 0$$

$$\therefore C \left(\frac{5}{4}, 0 \right)$$

Radius of circle is CP

$$\therefore r = \sqrt{\left(\frac{15}{4}\right)^2 + \left(\frac{15}{2}\right)^2}$$

$$= \sqrt{\frac{1125}{16}}$$

$$\therefore \text{Equation of circle} : \left(x - \frac{5}{4}\right)^2 + y^2 = \frac{1125}{16}$$

c) i) $xy = c^2$

differentiating, $y + x \frac{dy}{dx} = 0$

$$\frac{dy}{dx} = -\frac{y}{x}$$

At T $\left(ct, \frac{c}{t}\right)$

$$m_{\text{tangent}} = -\frac{1}{t^2}$$

E_{tangent} : $y - \frac{c}{t} = -\frac{1}{t^2} (x - ct)$

$$t^2 y - ct = x - ct$$

$$x + t^2 y = 2ct$$

ii) $x \text{ int} \Rightarrow y = 0$

$y \text{ int} \Rightarrow x = 0$

$$\therefore x = 2ct$$

$$t^2 y = 2ct$$

$$\therefore A(2ct, 0)$$

$$y = \frac{2c}{t}$$

Midpoint of AB = $\left(ct, \frac{c}{t}\right) = T$

$$B\left(0, \frac{2c}{t}\right)$$

$\angle BOA = 90^\circ \therefore AB$ is the diameter of a circle passing through O. T is the midpoint of this diameter and therefore the centre of the circle passing through O, A and B.



Question 14

$$a) \quad i) \quad \int_0^a f(x) dx = \int_0^a f(a-x) dx$$

$$\text{RHS} = \int_0^a f(a-x) dx$$

$$\text{let } u = a - x$$

$$du = -dx$$

$$= - \int_a^0 f(u) du$$

$$\text{when } x=0, u=a$$

$$x=a, u=0$$

$$= \int_0^a f(u) du$$

$$= \int_0^a f(x) dx$$

$$= \text{LHS}$$

$$ii) \quad \int_0^2 x(2-x)^5 dx$$

$$= \int_0^2 (2-x) x^5 dx$$

(from (i))

$$= \int_0^2 (2x^5 - x^6) dx$$

$$= \left[\frac{2x^6}{6} - \frac{x^7}{7} \right]_0^2$$

$$= \frac{64}{3} - \frac{128}{7}$$

$$= \frac{64}{21}$$



(13)

$$b) i) I_{2n+1} = \int_0^1 x^{2n+1} e^{x^2} dx$$

$$= \int_0^1 x^{2n} \cdot x e^{x^2} dx$$

$$= \left[\frac{1}{2} x^{2n} e^{x^2} \right]_0^1 - \int_0^1 \frac{1}{2} e^{x^2} \cdot 2n x^{2n-1} dx$$

$$= \frac{1}{2} e^1 - n \int_0^1 x^{2n-1} e^{x^2} dx$$

$$I_{2n+1} = \frac{1}{2} e - n I_{2n-1}$$

$$ii) I_5 = \frac{1}{2} e - 2 I_3$$

$$= \frac{1}{2} e - 2 \left(\frac{1}{2} e - I_1 \right)$$

$$= \frac{1}{2} e - e + 2 \int_0^1 x e^{x^2} dx$$

$$= -\frac{e}{2} + \left[e^{x^2} \right]_0^1$$

$$= -\frac{e}{2} + e^1 - e^0$$

$$= \frac{e}{2} - 1$$

$$9) i) (\cos \theta + i \sin \theta)^3$$

$$= \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$$

$$= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$$

ii) Using De Moivre's Thm,

$$(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$$

Equating real parts,

$$\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$$

$$= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta)$$



(14)

$$= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta$$

$$\cos 3\theta = 4 \cos^3 \theta - 3 \cos \theta$$

$$4 \cos^3 \theta = \cos 3\theta + 3 \cos \theta$$

$$\cos^3 \theta = \frac{1}{4} \cos 3\theta + \frac{3}{4} \cos \theta$$

$$\text{iii) } 4 \cos^3 \theta - 3 \cos \theta = 1$$

$$\therefore \cos 3\theta = 1 \quad (\text{from ci})$$

$$3\theta = 2k\pi$$

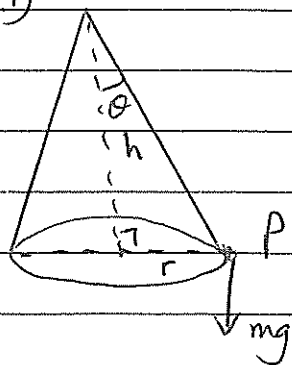
$$\theta = \frac{2k\pi}{3}$$

Smallest value occurs when $k=1$

$$\therefore \theta = \frac{2\pi}{3}$$

Question 15

a) i)



Resolving forces vertically & horizontally

$$T \cos \theta = mg \quad \text{--- (1)}$$

$$T \sin \theta = mr\omega^2 \quad \text{--- (2)}$$

$$(2) \div (1)$$

$$\frac{T \sin \theta}{T \cos \theta} = \frac{mr\omega^2}{mg}$$

$$\tan \theta = \frac{r\omega^2}{g}$$

$$\tan \theta = \frac{r}{h}$$

$$\therefore \frac{r\omega^2}{g} = \frac{r}{h}$$

$$\omega^2 = \frac{g}{h}$$

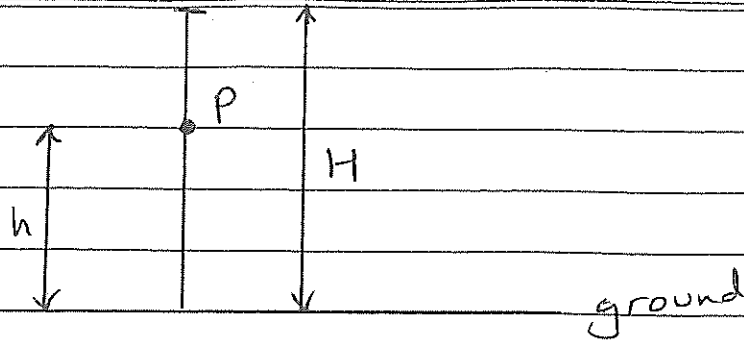
$$\omega = \sqrt{\frac{g}{h}}$$

$$\text{ii) Period} = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{g}{h}}} = \frac{2\pi\sqrt{h}}{\sqrt{g}}$$

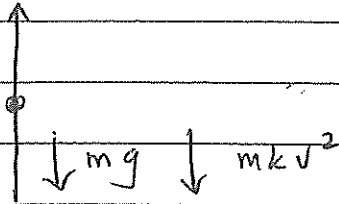


(15)

b)



i)



$$m\ddot{x} = -mg - mkv^2$$

$$\ddot{x} = -(g + kv^2)$$

$$v \frac{dv}{dx} = -(g + kv^2)$$

$$dx = \frac{-v}{g + kv^2} dv$$

$$\text{at } x = H, v = 0$$

$$x = 0, v = u$$

$$\therefore \int_0^H dx = -\frac{1}{2k} \int_u^0 \frac{2kv}{g + kv^2} dv$$

$$H = \frac{1}{2k} \int_0^u \frac{2kv}{g + kv^2} dv$$

$$= \frac{1}{2k} \left[\ln(g + kv^2) \right]_0^u$$

$$= \frac{1}{2k} \left(\ln(g + ku^2) - \ln g \right)$$

$$H = \frac{1}{2k} \ln \left(\frac{g + ku^2}{g} \right)$$



(16)

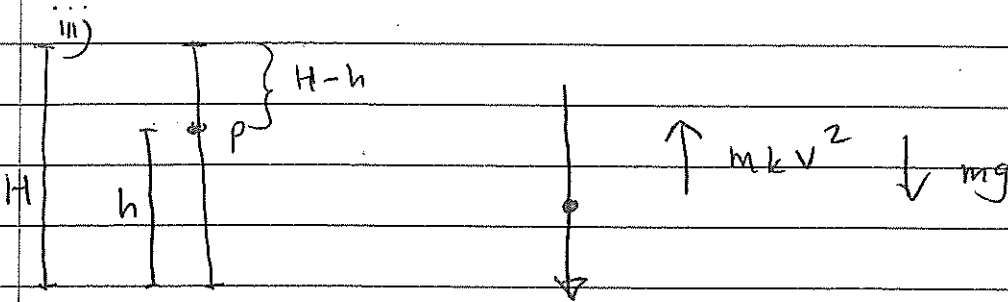
ii) when $x=h$, $v=V$

$$\int_0^h dx = -\frac{1}{2k} \int_u^V \frac{2kv}{g+kv^2} dv \quad (\text{from (i)})$$

$$h = \frac{1}{2k} \left[\ln(g+kv^2) \right]_V^u$$

$$= \frac{1}{2k} \left[\ln(g+ku^2) - \ln(g+kV^2) \right]$$

$$= \frac{1}{2k} \ln \left(\frac{g+ku^2}{g+kV^2} \right)$$



$$m\ddot{x} = mg - mkv^2$$

$$\ddot{x} = g - kv^2$$

$$v \frac{dv}{dx} = g - kv^2$$

$$dx = \frac{v}{g - kv^2} dv$$

$$\text{at } x=0, v=0$$

$$x=H-h, v=\frac{1}{2}V$$

$$\int_0^{H-h} dx = -\frac{1}{2k} \int_0^{\frac{V}{2}} \left(\frac{-2kv}{g - kv^2} \right) dv$$

$$H-h = -\frac{1}{2k} \left[\ln(g - kv^2) \right]_0^{\frac{V}{2}}$$

$$= -\frac{1}{2k} \left[\ln \left(g - k \frac{V^2}{4} \right) - \ln g \right] = -\frac{1}{2k} \ln \left(\frac{g - \frac{kV^2}{4}}{g} \right)$$

$$H-h = -\frac{1}{2k} \ln \left(\frac{4g - kV^2}{4g} \right) \quad \text{--- (1)}$$



(17)

Also, using H from (i) and h from (ii)

$$H-h = \frac{1}{2k} \ln \left(\frac{g+ku^2}{g} \right) - \frac{1}{2k} \ln \left(\frac{g+kv^2}{g} \right)$$

$$H-h = \frac{1}{2k} \ln \left(\frac{g+kv^2}{g} \right) \quad \text{--- (2)}$$

From (1) & (2)

$$\ln \left(\frac{g+kv^2}{g} \right) = -\ln \left(\frac{4g-kv^2}{4g} \right) = \ln \left(\frac{4g-kv^2}{4g} \right) = 1$$

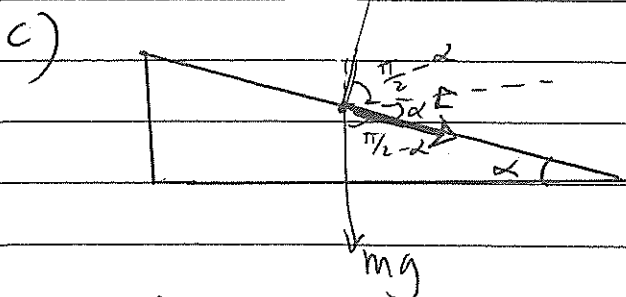
$$\frac{g+kv^2}{g} = \frac{4g}{4g-kv^2}$$

$$4g^2 + 4gkv^2 - gkv^2 - k^2v^4 = 4g^2$$

$$3gkv^2 = k^2v^4$$

$$v^2 = \frac{3g}{k}$$

$$v = \sqrt{\frac{3g}{k}}$$



Resolving Forces

Horizontally

$$\frac{mv^2}{r} = F \cos \alpha + N \cos \left(\frac{\pi}{2} - \alpha \right)$$

$$= F \cos \alpha + N \sin \alpha$$

$$\therefore F \cos \alpha = \frac{mv^2}{r} - N \sin \alpha \quad \text{--- (1)}$$

Vertically

$$N \cos \alpha - mg - F \cos \left(\frac{\pi}{2} - \alpha \right) = 0$$

$$N \cos \alpha - mg - F \sin \alpha = 0$$

$$\therefore F \sin \alpha = N \cos \alpha - mg$$

--- (2)



(18)

$$\text{ii) } \textcircled{1} \times \cos \alpha + \textcircled{2} \times \sin \alpha$$

$$F \cos^2 \alpha = \frac{mv^2}{r} \cos \alpha - N \sin \alpha \cos \alpha$$

+

$$F \sin^2 \alpha = N \sin \alpha \cos \alpha - mg \sin \alpha$$

$$= F (\cos^2 \alpha + \sin^2 \alpha) = \frac{mv^2}{r} \cos \alpha - mg \sin \alpha$$

$$F = \frac{m}{r} \cos \alpha \left(v^2 - \frac{gr \sin \alpha}{\cos \alpha} \right)$$

$$= \frac{m (v^2 - gr \tan \alpha)}{r} \cos \alpha$$

$$\text{iii) } r = 200 \text{ m}, g = 9.8 \text{ m/s}^2, F = 0$$

$$v = 100 \text{ km/h} = \frac{100 \times 1000}{3600} = \frac{250}{9} \text{ m/s}$$

$$\text{From (ii) } 0 = \frac{m (v^2 - gr \tan \alpha)}{r} \cos \alpha$$

$$v^2 - gr \tan \alpha = 0$$

$$\tan \alpha = \frac{v^2}{gr}$$

$$= \frac{(250/9)^2}{9.8 \times 200}$$

$$= 21.486$$

$$\alpha \approx 21^\circ 29'$$



(19)

Question 16

$$a) \quad P(x) = x^3(ax^2 + bx + c) = ax^5 + bx^4 + cx^3$$

$$\text{Let } Q(x) = P(x) - 1 = ax^5 + bx^4 + cx^3 - 1$$

Since $Q(x)$ is divisible by $(x-1)^3$, x is a triple root

$$Q'(x) = 5ax^4 + 4bx^3 + 3cx^2$$

$$Q'(1) = 5a + 4b + 3c = 0 \quad \text{--- (1)}$$

$$Q''(x) = 20ax^3 + 12bx^2 + 6cx$$

$$Q''(1) = 20a + 12b + 6c = 0 \quad \text{--- (2)}$$

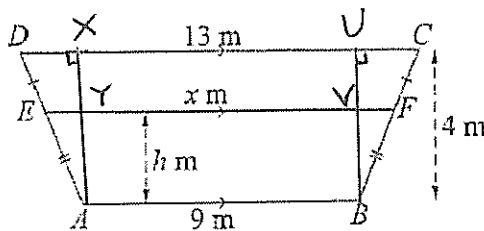
$$Q(1) = a + b + c = 0 \quad \text{--- (3)}$$

Solving (1), (2) & (3) simultaneously,

$$a = 6, \quad b = -15, \quad c = 10$$

$$\therefore P(x) = 6x^5 - 15x^4 + 10x^3$$

b)



Draw AX and $BV \perp DC$

$$\triangle AXD \equiv \triangle BVU \text{ (RHS)}$$

$$DX = CU = 2$$

$$\frac{EY}{AY} = \frac{DX}{AX} \quad \text{(ratio of matching sides of } \triangle \text{)}$$

$$\frac{EY}{h} = \frac{2}{4}$$

$$EY = \frac{h}{2} \quad ; \quad \text{Similarly, } VF = \frac{h}{2}$$

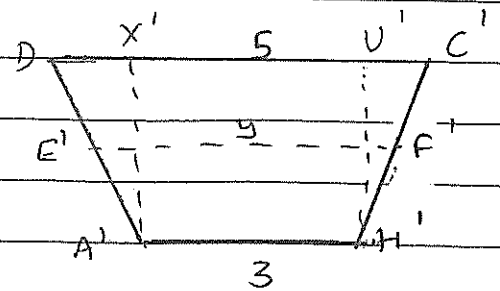
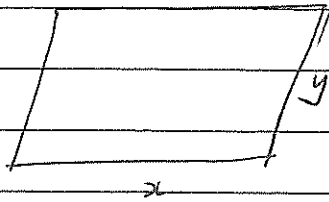
$$EF = EY + YV + VF$$

$$x = 9 + h$$



(20)

ii) Cross-sections // to the base will be rectangles



From (i) $x = 9 + h$

$D'X' = C'U' = 1$

Area of cross-section
 $= (9+h) \left(3 + \frac{h}{2}\right)$

$$= \frac{h^2}{2} + \frac{15h}{2} + 27$$

Using III Δs

$$\frac{E'Y'}{h} = \frac{1}{4}$$

$$E'Y' = \frac{h}{4}$$

$$y = 3 + 2 \times \frac{h}{4}$$

$$y = 3 + \frac{h}{2}$$

$$\Delta V = \lim_{h \rightarrow 0} \sum_{h=0}^4 \left(\frac{h^2}{2} + \frac{15h}{2} + 27 \right) \Delta h$$

$$V = \int_0^4 \left(\frac{h^2}{2} + \frac{15h}{2} + 27 \right) dh$$

$$= \left[\frac{h^3}{6} + \frac{15h^2}{4} + 27h \right]_0^4$$

$$V = 178 \frac{2}{3} \text{ m}^3$$

(21)

c) i) $y = mx + k$ — (1)

$xy = c^2$ — (2)

Solving (1) & (2)

$$x(mx + k) = c^2$$

$$mx^2 + kx = c^2$$

$$mx^2 + kx - c^2 = 0$$

$$\Delta = k^2 - 4mc^2$$

Since $y = mx + k$ is tangent $\Delta = 0$

$$\therefore k^2 - 4mc^2 = 0$$

ii) The equation of the line through $(-1, -3)$ with gradient m is:

$$y + 3 = m(x + 1)$$

$$y = mx + m - 3$$

$$y = mx + k \quad \text{where } k = m - 3.$$

This is tangent to $xy = 4$ if $k^2 + 16m = 0$

$$(m - 3)^2 + 16m = 0$$

$$m^2 + 10m + 9 = 0$$

$$(m + 9)(m + 1) = 0$$

$$m = -9 \text{ or } m = -1$$

Equation of tangent $y = mx + m - 3$

$$m = -9; \quad y = -9x - 12$$

$$m = -1; \quad y = -x - 4$$

Solving $y = -9x - 12$ and $xy = 4 \Rightarrow x(-9x - 12) = 4$

Solving $y = -x - 4$ and $xy = 4$,

$$x(-x - 4) = 4$$

$$x^2 + 4x + 4 = 0$$

$$(x + 2)^2 = 0$$

$$x = -2, \quad y = -2 \Rightarrow (-2, -2)$$

$$9x^2 + 12x + 4 = 0$$

$$(3x + 2)^2 = 0$$

$$x = -\frac{2}{3}, \quad y = -6$$

$$\left(-\frac{2}{3}, -6\right)$$